



Chapter 1 – Physics and Measurements

Standards of Length, mass and time

Dimensional Analysis

Conversion of units

Significant Figures

Measurements

- ▶ Used to describe natural phenomena
- ▶ Each measurement is associated with
 - a physical quantity
 - a unit

Quantity	SI Unit
Length	meter
Mass	kilogram
Time	second
Temperature	Kelvin
Electric Current	Ampere
Luminous Intensity	Candela
Amount of Substance	mole

Section 1.1

Scientific Notation

- ▶ Writing large numbers in groups of three (starting from the right hand side)
 - No commas
 - Standard international notation
- ▶ Examples:
 - 25 100
 - 5.123 456 789 12

Section 1.1

Prefixes

- ▶ “Powers of 10”
- ▶ Prefixes correspond to powers of 10.
- ▶ Each prefix has a specific name.
- ▶ Each prefix has a specific abbreviation.
- ▶ They are multipliers of the basic unit.
- ▶ Examples:
 - 1 mm = 10^{-3} m
 - 1 mg = 10^{-3} g
 - 1 Gs = 10^9 s
 - 1TB = 10^{12} B

Section 1.1

Prefixes, cont.

TABLE 1.4 *Prefixes for Powers of Ten*

Power	Prefix	Abbreviation	Power	Prefix	Abbreviation
10^{-24}	yocto	y	10^3	kilo	k
10^{-21}	zepto	z	10^6	mega	M
10^{-18}	atto	a	10^9	giga	G
10^{-15}	femto	f	10^{12}	tera	T
10^{-12}	pico	p	10^{15}	peta	P
10^{-9}	nano	n	10^{18}	exa	E
10^{-6}	micro	μ	10^{21}	zetta	Z
10^{-3}	milli	m	10^{24}	yotta	Y
10^{-2}	centi	c			
10^{-1}	deci	d			

Section 1.1

Examples

- ▶ Mass of the Earth: $5.98 \times 10^{24} \text{ kg} = 5.98 \times 10^3 \text{ Yg}$
- ▶ Period of visible light: $1 \times 10^{-15} \text{ s} = 1 \text{ fs}$
- ▶ Mean radius of the Earth: $6.37 \times 10^6 \text{ m} = 6.37 \text{ Mm}$

Section 1.1

Basic Quantities and Their Dimension

- ▶ Dimensions refer to the basic type of measurement unit / physical quantity
- ▶ Dimensions denoted with square brackets.
 - Length [L]
 - Mass [M]
 - Time [T]
- ▶ Each dimension can have different units. E.g. length in meters, miles, inches, cubits

Section 1.3

Dimensions and Units

TABLE 1.5 *Dimensions and Units of Four Derived Quantities*

Quantity	Area (A)	Volume (V)	Speed (v)	Acceleration (a)
Dimensions	L^2	L^3	L/T	L/T^2
SI units	m^2	m^3	m/s	m/s^2
U.S. customary units	ft^2	ft^3	ft/s	ft/s^2

Section 1.3

Fundamental and Derived Units

- ▶ Derived quantities can be expressed as a mathematical combination of fundamental quantities [L], [T], [M].

- Area: $[A] \equiv [L] \times [L]$ m^2
- Speed: $[v] \equiv [\text{length}] / [\text{time}] \equiv [L] / [T]$ m/s
- Density: $[\rho] \equiv \frac{[\text{mass}]}{[\text{volume}]} \equiv \frac{[M]}{[L] \times [L] \times [L]}$ kg/m^3
- Force: $[F] \equiv [\text{mass}] \times [\text{acceleration}]$ $\text{kg} \cdot \text{m} \cdot \text{s}^{-2} \equiv \text{N}$
 $\equiv [M] \times \frac{[L]}{[T] \times [T]}$

Section 1.1 & 1.3

Dimensional Analysis

- ▶ Technique to check the correctness of an equation or to assist in deriving an equation
- ▶ Dimensions (length, mass, time, combinations) can be treated as algebraic quantities.
 - Add, subtract, multiply, divide
- ▶ Both sides of equation must have the same dimensions.
- ▶ We can only add, subtract or set equal to one another quantities that have the same dimensions

Section 1.3

Dimensional Analysis: example

- ▶ Given the equation: $x = \frac{1}{2}at^2$
- ▶ Check dimensions on each side:

$$L = \frac{L}{T^2} \cdot T^2 = L$$

- ▶ The T^2 's cancel out, leaving L for the dimensions of each side.
 - The equation is dimensionally correct.
 - There are no dimensions for the constant.

Section 1.3

Symbols

- ▶ The symbol used in an equation is not necessarily the symbol used for its dimension.
- ▶ Some quantities have one symbol used consistently.
 - For example, time is usually represented by t .
- ▶ Some quantities have many symbols used, depending upon the specific situation.
 - For example, lengths may be x, y, z, r, d, h , etc.

Section 1.3

Conversion of Units

- ▶ All measurements have a unit associated with it
- ▶ When units are not consistent, you may need to convert to appropriate ones.
- ▶ See Appendix A for an extensive list of conversion factors.
- ▶ Units can be treated like algebraic quantities that can cancel each other out.

Section 1.4

Conversion

- ▶ **Always** include units for every quantity, and carry the units through the entire calculation.
 - Will help detect possible errors
- ▶ **Multiply original value by a ratio equal to one.**
- ▶ Example:

$$15.0 \text{ in} = ? \text{ cm}$$

$$15.0 \text{ in} \left(\frac{2.54 \text{ cm}}{1 \text{ in}} \right) = 38.1 \text{ cm}$$

- Note the value inside the parentheses is equal to 1, since 1 inch is defined as 2.54 cm.

Section 1.4

Example

How many milliseconds are there in 1 kilosecond?

Section 1.4

Example

The density of lead is 11.35 g/cm^3 . If we have a bar of gold with dimensions $250 \text{ mm} \times 35 \text{ mm} \times 70 \text{ mm}$ and it weighs 11.8 kg , is the density of gold more or less than that of lead?

Section 1.4

Example

The fastest growing plant on record is a *Hesperoyucca whipplei* that grew 3.7 m in 14 days. What was its growth rate in micrometers per second?

Section 1.4

Uncertainty in Measurements

- ▶ There is uncertainty in every measurement – this uncertainty carries over through the calculations.
 - May be due to the apparatus, the experimenter, and/or the number of measurements made
 - Need a technique to account for this uncertainty
- ▶ Ruler vs vernier caliper

Section 1.6

Significant Figures

- ▶ A significant figure is one that is reliably known.
- ▶ Zeros may or may not be significant.
 - Those used to position the decimal point are not significant. E.g. 0.0075 m and 0.00075 m both have 2 significant figures
 - To remove ambiguity, use scientific notation. E.g. 7.5×10^{-3} m and 7.5×10^{-4} m
- ▶ In a measurement, the significant figures include the first estimated digit.

Section 1.6

Significant Figures examples

- ▶ 10.0 m has 3 significant figures
 - The decimal point gives information about the reliability of the measurement.
- ▶ 1500 m is ambiguous
 - Use 1.5×10^3 m for 2 significant figures
 - Use 1.50×10^3 m for 3 significant figures
 - Use 1.500×10^3 m for 4 significant figures

Section 1.6

Operations with Significant Figures – Multiplying or Dividing

- ▶ When multiplying or dividing several quantities, the number of significant figures in the final answer is the same as the number of significant figures in the quantity having the smallest number of significant figures.
- ▶ Example: $25.57 \text{ m} \times 2.45 \text{ m} = 62.6 \text{ m}^2$
 - The 2.45 m limits your result to 3 significant figures.

Section 1.6

Operations with Significant Figures – Adding or Subtracting

- ▶ When adding or subtracting, the number of decimal places in the result should equal the smallest number of decimal places in any term in the sum or difference.
- ▶ Example: $135 \text{ cm} + 3.25 \text{ cm} = 138 \text{ cm}$
 - The 135 cm limits your answer to the units decimal value.

Section 1.6

Rounding

- ▶ Round up
- ▶ Round down
- ▶ Saving rounding until the final result will help eliminate accumulation of errors.
- ▶ It is useful to perform the solution in algebraic form and wait until the end to enter numerical values.

Section 1.6